EMERGING TECHNOLOGIES: AN OVERVIEW AND ANALYSIS OF ITS AREAS

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1. Introduction

Emerging technologies have the potential to rapidly change how people live and conduct business, but they are still in their infancy and at the beginning stages of development and evolution. Technologies such as artificial intelligence have been around for decades (since the 1950s) and are still classified as “emerging” because they keep evolving, find new applications, improve their performance, and are still a subject of research. Thus, the term emergence is less about how novel a technology is and more about its stage of development and its potential for ongoing growth and impact. This notion leads to the key features characterizing emerging technologies.

Understanding how technologies emerge is essential for organizations that want to remain competitive, as well as for individuals — particularly IT professionals — who will be expected to work with, adapt to, and eventually implement these tools in real-world settings. These needs also make it important to examine the characteristics that distinguish emerging technologies.

2. Characteristics of Emerging Technologies

According to Rotolo, Hicks, and Martin (2015), a technology can be classified as "emerging" based on five (5) defining characteristics rather than simply its age. These characteristics worked together to identify true emerging technologies from technologies that are merely new.

Table 1.The Five Characteristics of Emerging Technologies

Characteristic	Description	Example
Radical Novelty	The technology introduces a genuinely new approach or applies an existing idea in a new way, rather than just improving something old. It solves problems others did not try.	Artificial intelligence is applied to an ever-growing number of tasks in fundamentally different ways every time
Relatively Fast Growth	The technology expands quickly — as evidenced by rising investment, more people/organizations involved, and increase of research outputs.	The number of companies, universities, and individuals adopting AI continue to grow rapidly.
Coherence	Different fields or technologies begin to merge and work together for a unified purpose.	Cloud computing, artificial intelligence, data analytics and robotics combined to serve shared goals.
Prominent Impact	The technology provides wide-reaching effects across society and different industries / fields, not just within a narrow niche.	AI spreads rapidly to different sectors (i.e business, healthcare, academe, etc.)
Uncertainty and Unambiguity	The reason for its full impact lies in the future, there is no guarantee how the technology will be used.	Various applications are still malleable, still uncertain if it may or may not be successful.

3. Emerging Technology Areas

The World Economic Forum identifies the different technology areas that have enough chance of changing the old practices in the future. Module 1 highlights four of these areas in particular: Augmented Reality (AR), Artificial Intelligence (AI), Machine Learning, and Quantum Computing.

3.1 Augmented Reality (AR)

Rather than replacing what a person sees, Augmented Reality (AR) overlays digital content — images, videos, audio, etc. — directly onto the physical world. It combines both the physical and digital worlds to create something unique and spectacular. The technology can be seen in applications such as Pokémon GO, in-car displays that assist drivers while parking, etc. Augmented Reality (AR) is expected to grow rapidly and is considered a core part of the Fourth Industrial Revolution.

3.2 Artificial Intelligence (AI)

AI is a technology where it is built to imitate human thinking. It includes learning, reasoning,and self-correction.

3.3 Machine Learning

Machine learning is a branch of AI that allows systems to learn from data and experience, similarly to how a human learns. It grew rapidly to different fields such as

healthcare, and financial and marketing services. In example, in healthcare, machine learning helps researchers identify and develop new medicines faster than the old approach.

3.4 Quantum Computing

Quantum computing is a special type of computer that uses principles of quantum mechanics to solve problems that regular computers struggle with. Instead of using the standard bits (0 or 1), they use qubits that represent both 0 and 1 at the same time. Companies like IBM, Google, Rigetti Computing, and IonQ are adapting to the development of quantum computing.

4. The Industrial Revolution

The Industrial revolution classified major shifts on how products/goods were produced. There have been four (4) stages so far:

Table 2. Stages of the Industrial Revolution

Stages	Time Period	Key Development
Industry 1.0	18th Century	Mechanical production using steam and hydro power
Industry 2.0	19th Century	Electricity made possible to produce larger-scale, more advanced manufacturing
Industry 3.0	20th Century (Computer era)	Electronics and IT automated production further
Industry 4.0	Present Day	Smart technologies (i.e IoT, AI, AR, etc.) change how people live and work

5. Building Blocks of Industry 4.0

- **Big Data Analytics -** Collecting and analyzing large volumes of data to identify patterns and support real-time business decisions.
- **Internet of Things (IoT) -** Everyday devices connected to the internet that can communicate and respond in real time.
- **Cloud Computing -** Delivering computing services (i.e storage, software, etc.) over the internet instead of in local hardware.
- **Augmented Reality -** This technology is used in industrial settings to help workers, such as helping locate warehouse items or providing repair instructions on mobile devices.

6. Conclusion

Emerging technologies such as AI, AR, machine learning,and quantum computing are reshaping the industries as part of the Fourth Industrial Revolution. Even if they bring major benefits in efficiency, innovation, and decision-making, they also come with real risks — like job disruption and uncertainty about long-term outcomes. Grasping these technologies is essential for future IT professionals who will need to adapt to and work within this changing landscape.

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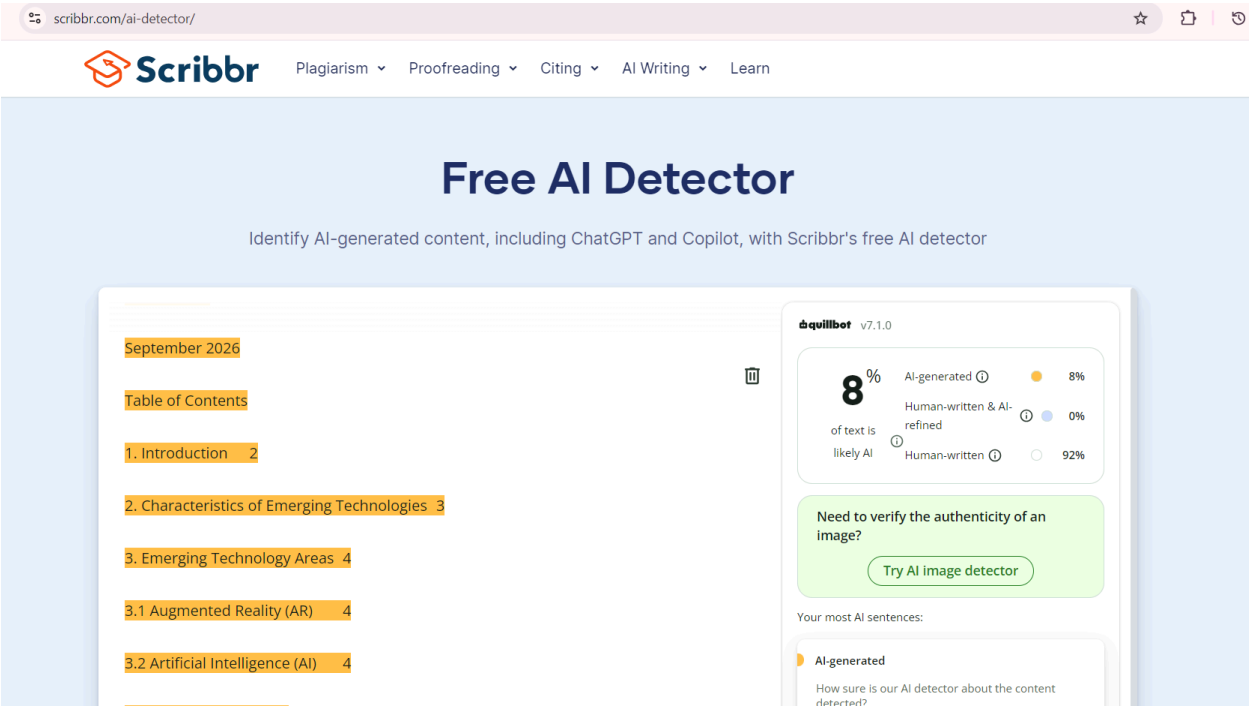
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8. AI Checker Result

Tool Used: Scribbr

Percent: 8%



9. Grammarly Result

Percent: 99%

